The Neglected Work Problem

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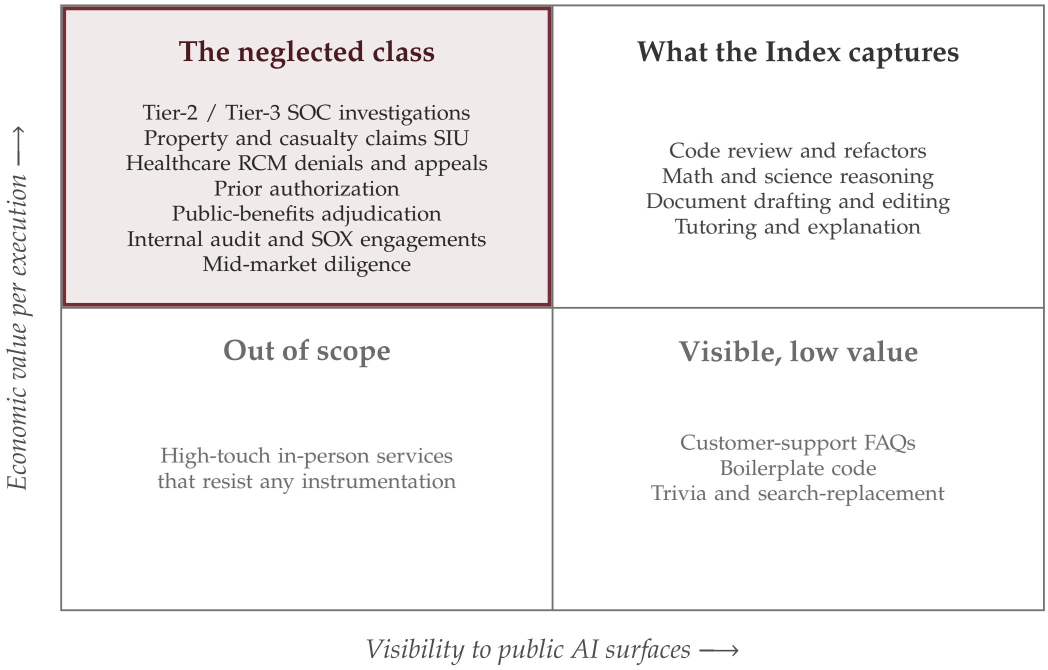


Figure 1: Where the Anthropic Economic Index measures, and where it does not. The highlighted quadrant is high-value work that is invisible to public AI surfaces because of customer contracts, regulation, and tooling. That is the class top RL environment companies neglect.

Anthropic's Economic Index is the best public artifact for tracking AI displacement, and it is wrong in a way that matters. It indexes what users bring to Claude.ai, prices it against May 2024 BLS wages, and has Claude judge whether Claude succeeded. The work that creates the largest pool of remaining economic value, long-horizon process work inside regulated services firms, never appears in that distribution and cannot. The same gap is why top RL environment companies miss this class of work, and it is the opening I want to write toward.

What the Index actually measures

The January 2026 Economic Primitives paper and the March 2026 Learning Curves follow-up share a stack. About a million Claude.ai conversations and a million API conversations are classified against the O*NET occupational taxonomy, weighted by BLS Occupational Employment and Wage Statistics from May 2024, and assigned five primitives (task complexity, human and AI skills, use case, AI autonomy, task success) by Claude grading the conversations. The authors flag some of their own limits in passing. Three of those are load-bearing for the rest of this essay.

Claude.ai usage is downstream of whether AI seems likely to help, on a surface the employer permits. A Tier-2 analyst running a 90 minute investigation across an enterprise security stack is not pasting alert payloads into a chatbot. The work is not low-value. The analyst's firm signed a contract with the customer that forbids pasting telemetry into a public surface. The Index cannot see that conversation, by design.

Wages are not the marginal economic value of the underlying work. BLS OEWS reports equilibrium wages under current labor market conditions, which reflect credentialing, geographic supply, union power, and outsourcing. A Tier-2 investigation done by an offshore analyst at a contracted hourly cost is the same operational work as a US Information Security Analyst whose

May 2024 median wage was $124,910. The Index assigns value at the US occupation code and undercounts the offshore-contracted variant. The most AI-displaceable seats in the world get attributed at the wage of the salaried seat they replaced a decade ago.

Success is Claude grading Claude. The fifth primitive has no external ground truth, so in any domain where success cannot be checked from outside (most operational work), the Index reduces to "Claude thinks Claude did X." This is the same problem you have flagged on X about the human data market, where vendors pay for experts they themselves are not equipped to grade. Anthropic's version is worse, because the rater and the rated are the same model.

The Index is rigorous as a longitudinal tracker of Claude.ai product penetration among English-speaking knowledge workers. The error is reading it as a map of where AI creates economic value. It is a map of where AI visibly creates economic value to Anthropic.

The neglected class

A class of work is in the neglected set if it meets all five of: long-horizon (median execution at least 30 minutes, at least 10 tool calls), multi-stakeholder (at least three humans with differing incentives touch a single execution), strictly unverifiable (no scalar ground truth at the trajectory level, only process quality and rare-event tail outcomes), regulated or compliance-mediated, and services-mediated (done predominantly by managed services firms, BPO, third-party administrators, RCM vendors, audit shops, or consultancies rather than in-house labor). This excludes most code review, most legal contract work, most customer support, most healthcare diagnosis. The industries that pass all five are the highlighted quadrant in Figure 1.

I worked Tier-3 in the UNC Pembroke SOC from July 2023 to May 2026, building Cisco XDR automations and Python triage enrichment. The work that the Index cannot see and that human-data vendors cannot procure is the kind of trajectory you find in any mid-market managed-security operation. A representative one runs roughly like this. A DNS-resolution alert fires on a host reaching a high-entropy domain. The analyst pivots into the SIEM, looks at process-creation events on the host, finds a parent shell process spawning the resolution. Pivots into endpoint detection, sees a benign-looking productivity-app parent with an anomalous macro pattern. Pivots for the file hash to a reputation service, gets a handful of generic detections. Pivots into network-flow data to check lateral movement. Decides to escalate.

That trajectory runs 45 to 90 minutes in the median with a long tail, and touches a handful of tools across multiple vendors. Three stakeholders judge it: the incident-response lead, the operations manager, and the customer's security owner, each with a different reward function. There is no scalar ground truth; the only hard one (was there a real incident) fires only in the long tail. The whole thing lives behind a customer-contract perimeter, and the data cannot leave the SIEM for Claude.ai or be shipped to Scale or Mercor.

Global managed security services revenue is around $43B in 2026 and SIEM is another $11B. By workflow-value logic, measured in breach loss, regulatory penalty, and customer churn, this is one of the highest-leverage operational functions in the economy. My read is the Index attributes a tiny fraction of total value to it. The same five-criterion test applied to claims SIU, RCM denials, or internal-audit work produces the same gap.

Top RL environment companies miss this class for structural reasons, not laziness. Customer concentration drives roadmap, as you have written in "Kings and Electors" and "New Age Commodities": env vendors get most of their revenue from labs, labs ask for verifiable domains first (code, math, science) because the reward signal is cheap, so unverifiable long-horizon ops is phase two and phase two never comes. The contractor model fits credentialed individuals, not seat-time operators. Scale, Mercor, Micro1, Invisible, and Surge source a PhD chemist, a JD, an MD; their primitive cannot procure a working managed-security shift floor with three months of historical investigations. The telephone game compounds. By the time procurement reaches operational data it has passed through sales contact, customer procurement, a security-cleared liaison, a sanitization pipeline the vendor did not design, and a labeling team that has never worked a shift. What arrives is contrived. Process-reward and long-horizon-reward design for rare-event trajectories is open research, and env vendors staff for delivery. Synthetic data does not bridge it, because synthetic trajectories inherit the model's blind spots and no frontier model has worked an operational shift.

Procurement: equity-for-instrumentation

Procurement is the binding constraint. The mechanism has to satisfy three parties at once: the services firm whose workflows get instrumented, the end customer whose data lives in those workflows, and the lab that funds the research. Pay-per-task and pay-per-hour fail all three. The mechanism I propose is equity-for-instrumentation paired with a privacy-preserving capture pipeline. No raw data ever leaves the firm.

Mid-market services firms are mostly private, margin-compressed, and looking for any defensible AI angle that is not buying ChatGPT seats. Handing their data to a broker violates customer contracts and erodes their moat. Renting analysts by the hour misses the value, because the value is not in labels but in the operational context the analyst inhabits. The structure I would write: the services firm grants exclusive instrumentation rights to its workflows, subject to customer consent and on-prem redaction. In exchange it receives equity warrants in the venture, revenue share on derived model artifacts, and co-developed fine-tuned tooling it can deploy with its own customers without losing moat (exact terms negotiated per pilot). The end customer gets an opt-in: redacted aggregated workflow telemetry leaves the perimeter only after consent and redaction, in exchange for early access to that tooling, with operational metric improvement (mean time to respond, denial rate, loss ratio) measurable against their pre-instrumentation baseline. The lab funds the venture in exchange for trained artifacts and curated bundles, not raw data, which keeps the firm's moat intact and removes the customer-consent risk from the lab.

Vendor model	What they procure	Pricing primitive	Why it misses ops data
Scale / Surge	Labels on lab-supplied raw data	Per-label, per-hour	No access to live workflows, no operational context
Mercor / Micro1	Credentialed individuals on demand	Per-expert-hour	Cannot capture seat-time process orchestration
Plato / Hud / Halluminate	Synthesized environments	Per-env or seat license	Synthetic blind spots on workflows no model has seen
This proposal	*Instrumentation rights to live workflows*	*Equity + revenue share on artifacts*	*Aligns with the firm's moat, captures process trajectories under contract*

Table 1: Existing vendor models against the equity-for-instrumentation mechanism.

Raw operational workflows are messy in a specific way: recorded across tools the vendor does not own, behind customer-contract perimeters the vendor cannot cross, with redaction requirements the vendor cannot audit from outside. Figure 2 shows the five stages that turn that mess into fine-tuning-amenable bundles. Every stage uses off-the-shelf components.

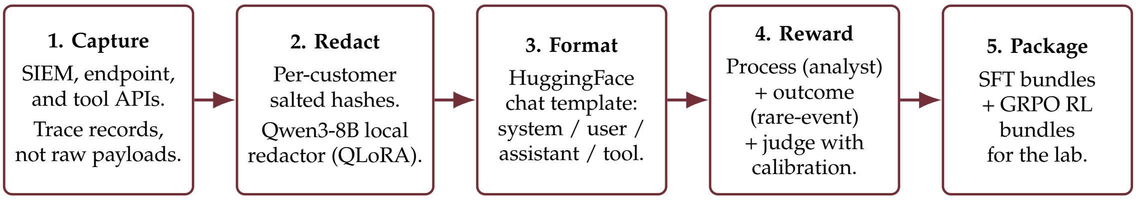
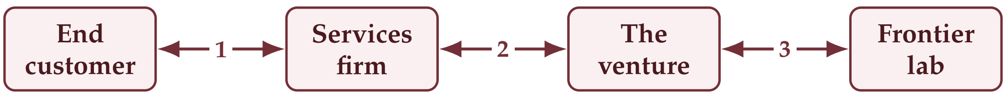


Figure 2: Five-stage capture-and-map pipeline. Stages 1 to 4 run inside the services firm's perimeter or against firm-owned outputs; only Stage 5 bundles leave the firm, and only the lab pays for them.

Stage 1. An agent inside the firm's perimeter, hooked into existing tooling APIs, records timestamp, actor, tool call, input, output, intermediate artifact, and decision at process-tree granularity. *Stage 2.* Deterministic salted hashes for customer identifiers, plus a Qwen3-8B local redactor fine-tuned with QLoRA on the firm's historical incidents (same stack as DPBench, transferred to a redaction objective); redaction errors escalate to a reviewer inside the firm, never to the venture. *Stage 3.* Trajectories serialized into HuggingFace chat templates so every modern post-training stack consumes them natively. *Stage 4.* Three reward streams per trajectory: process rewards labeled by senior analysts against the firm's playbook (two reviewers per checkpoint, Cohen's κ audited monthly); outcome rewards backfilled from the rare-event tail; an LLM judge for steps where process labeling is too expensive, with judge agreement against the human-labeled set audited on a small calibration sample. That calibration sample is the antidote to the reward-hacking failure mode DPBench surfaced, where judges look fine in aggregate but hide tail breakdowns. *Stage 5.* SFT bundles from Stage 3; RL bundles combining Stage 3 with Stage 4 dense process rewards and sparse outcome rewards, weighted with GRPO group-relative advantages.



1 Customer's opt-in workflow contribution in exchange for the firm's operational metric improvement (mean time to respond, denial rate, loss ratio).

2 Firm's instrumentation rights in exchange for the venture's equity warrants, revenue share, and co-developed fine-tuned tooling.

3 Venture's trained artifacts and curated bundles in exchange for the lab's research funding.

Figure 3: Linear stakeholder flow. Each numbered exchange is a sustainable two-way trade: the firm gets a moat instead of a one-time payment, the customer gets measurable operational improvement, the lab gets research artifacts no incumbent vendor can deliver.

Execution plan: Steps 1, 2, 3

What I would do, starting Monday of week one.

Step 1. Sign one services firm to a 90-day pilot and capture 500 trajectories. Without a real dataset under a real contract every next step is a deck. Build a target list of around forty mid-market managed-security firms anchored on regulated-industry customers in healthcare or financial services, sourced from Gartner peer reviews, AWS managed-security partner lists, NCSC accredited lists, and my own SOC network. Cold outreach is grounded in shared vocabulary, not a sales pitch: I can walk into a conversation with the founder of a managed-security firm and discuss XDR automation pain points within two minutes. Term sheet template uses equity warrants, revenue share on derived artifacts, a vertical exclusivity window, and mutual termination on customer-consent failure (specific percentages negotiated per pilot). Privacy counsel engaged before the first conversation, not after. Target: one signed pilot in six weeks, 500 captured trajectories by week twelve.

Step 2. Ship a workflow-faithful eval and prove the gap before training anything. Carve a

50-trajectory eval out of the captured 500. Each trajectory becomes a multi-turn agent benchmark with the same tool catalog, same alert, same playbook, scored on process-reward agreement with senior analysts and outcome-reward agreement on the rare-event tail. Run it on Claude Opus 4.7, GPT-5.5, Gemini 3.1 Pro, plus open-weights baselines (Qwen3-14B, GLM 5.1), both zero-shot and with workflow-specific tool-use prompts. Compare against the same models' public agentic benchmarks (SWE-bench Pro, GAIA, τ-bench, OSWorld, Terminal-Bench 2.0). The interesting result is the gap: where do public benchmarks predict workflow performance, and where do they fail. Publish the methodology (not the data) as a short report or arXiv preprint. This is publication-grade work and a credible artifact for lab BD conversations, and it extends my existing publication line on multi-agent coordination and LLM eval under adversarial and multilingual conditions.

Step 3. Post-train a vertical model and prove the slope. Take the 500 trajectories, run them through the pipeline, post-train Qwen3-14B (or the best open-weights base at runtime) with SFT on Stage 3 plus GRPO with Stage 4 process and outcome rewards, then re-run the Step 2 eval. A single GRPO run on a 14B base with 500 trajectories sits comfortably inside a one-person compute budget on rented H100s. Two ablations baked in: SFT-only versus SFT plus RL, and process-only versus process plus outcome. If the fine-tuned 14B beats frontier closed-source models on the workflow-faithful eval by at least ten points on process reward (and prior vertical-RL work suggests it will), the conversation with a lab BD team becomes a different conversation. The artifact you are bringing them is a capability demonstration they cannot replicate without partnering on the data side, not a data dump. If Step 3 lands, the next move is mechanical: replicate the playbook in claims SIU, RCM denials, and internal-audit engagements with the same contract template and the same pipeline. The defensibility is in the mechanism, not any single vertical, and the mechanism is what no incumbent vendor will copy because it requires giving up the labor-arbitrage model their existing investors priced.

Your closing line in "New Age Commodities" is that specialized vendors will fragment AI infrastructure markets rather than consolidate around generalist incumbents. This is the largest fragment still unpicked. It is unpicked because the procurement primitive that exists today (rent a credentialed individual for two hours) does not fit the shape of the work. The primitive that does is a new business model, not a new product feature.